

Problem Formalization

INPUTS



Transaction Graphs

$$\{G_k=(V_k,E_k,X_k)\}_{k=1}^K$$

K chains, $X_k \in \mathbb{R}^{|V| \times d}$



Network States

$$\{S_i=(\lambda_i,\beta_i,\tau_i)\}_{i=1}^n$$

n nodes: latency, BW, τ

SYSTEM CONSTRAINTS



C1: Safety

$$\forall i,j \in H: \hat{y}_i(v) = \hat{y}_j(v)$$

Honest nodes produce identical classifications



C2: Liveness

$$\forall v: \exists t < T_{\max}, \hat{y}(v) \text{ decided}$$

All transactions classified within bounded time



C3: Accuracy

$$F1(\hat{y}, y) \geq \theta_{\text{acc}}$$

Detection quality meets threshold (e.g., $F1 \geq 85\%$)



C4: Efficiency

$$O(\text{comm}) \leq O(nd)$$

Linear communication complexity per round

OUTPUT



Consistent Anomaly Classification

$$\hat{y}: V \rightarrow \{\text{illicit, legitimate}\}, \forall i,j \in H: \hat{y}_i \equiv \hat{y}_j$$

ADVERSARIAL CHALLENGES



Byzantine Adversaries

$$|B| = f < n/3$$

Arbitrary behavior



Network Heterogeneity

$$\lambda_i, \beta_i, \tau_i \sim \text{varied}$$

Non-IID

Notation: H =honest, B =Byzantine, $n=|H|+|B|$, d =feature dim, K =chains